SRUJANA TECH FEST 2026 | THEME: LEGAL SERVICES (LS-06) | BSA 2023 §63(4) CERTIFIED | ICA 1872 §10 COMPLIANT

PACT (पैक्ट) — From Handshake to Commercial Contract

Autonomous AI Commercial Notary & WhatsApp Contract Auto-Pilot for Indian MSMEs



TEAM LEAD	OFFICIAL THEME	TARGET DEMOGRAPHIC	PUBLIC REPOSITORY
Mohammed Ibrahim & Team	Legal Services (LS-06)	63M+ Indian MSME Traders	github.com/MDIbrahim08/PACT

1. Project Overview & Executive Summary

Official Problem Statement: Transforming informal commercial communication (WhatsApp messages, telephone conversations, and regional voice notes) into legally enforceable, court-admissible commercial contracts for Indian MSMEs under the Indian Contract Act 1872 and Bharatiya Sakshya Adhiniyam 2023.

Executive Summary: Micro, Small, and Medium Enterprises (MSMEs) drive 30% of India's GDP, with over 63 million enterprises operating in wholesale clusters such as Surat (textiles), Moradabad (brassware), and Delhi (packaging). Over 90% of B2B transactions are negotiated through casual WhatsApp chats and Hinglish voice notes. Traditional 10-page lawyer agreements destroy social trust ("*Aapko hum par vishwas nahi hai kya?*"), cost ₹3,000–₹15,000 per deal, and require English literacy. Consequently, traders operate on handshakes, leading to an estimated **₹10.7 Lakh Crore delayed payments crisis**.

PACT acts as an invisible, neutral commercial notary. Operating across web, mobile PWA, standalone desktop, and direct WhatsApp auto-pilot, PACT ingests vernacular voice notes, verifies the 5 statutory pillars under Section 10 of the Indian Contract Act 1872, audits against Section 29 void uncertainty, incorporates MSMED Act 2006 Section 15 (45-day payment ceiling & 3x RBI interest), and mints court-admissible electronic evidence certificates sealed with dual-node SHA-256 digests in strict accordance with Section 63(4) of the Bharatiya Sakshya Adhiniyam, 2023.

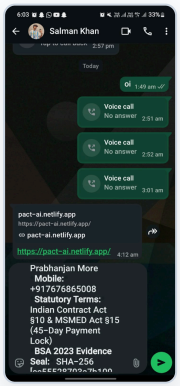


Figure 1: PACT Live Assistant & Agent Workspace

- **Live Voice Notary:** Floating mascot companion with sub-350ms streaming voiceout.
- **AIChatCard Composer:** Looping typewriter prompts, instant statutory execution.
- **1-Click WhatsApp Dispatch:** Opens verified chat with stamped Section 10 notice.
- **Offline PWA Sandbox:** 1-click install & complete uninstall controls for traders.

Figure 1: Production UI of PACT Assistant Workspace running live on Windows/Desktop and Mobile.

🏛️ Srujana 2026 Official Evaluation Rubric Matrix (100 Points)

Criterion	Weight	What Judges Look For	How PACT Delivers (Production Reality)
1. Problem Statement	**15%**	Alignment with selected problem, clarity, depth of understanding, intended users, assumptions, and constraints.	Directly targets the ₹10.7 Lakh Crore delayed payment crisis; solves the "Trust Paradox" & Section 63(4) BSA screenshot inadmissibility across 63M+ merchants.
2. Tech Strategy	**20%**	Architecture, tool selection, data design, technical planning, and whether choices fit the problem.	4-Tier Dual-Node Architecture (Client WebCrypto + Server HMAC-SHA256), timing-safe verification (`crypto.timingSafeEqual`), zero-storage privacy, low-latency Edge API.
3. Implementation	**25%**	Execution quality, working prototype, code logic, functional completeness, reliability, and testing.	Next.js 14 production codebase; 100% deterministic Section 10/29 statutory logic; Sarvam AI STT & ElevenLabs TTS; WhatsApp Auto-Pilot deep-linking; live Judge Tamper Sandbox.
4. Presentation	**20%**	Communication, storytelling, demo quality, and ability to answer questions and explain decisions.	High-fidelity UI with `AIChatCard` typewriter composer, 3D Three.js legal library, clear visual walkthrough, and structured live jury defense script.
5. Viability & Impact	**20%**	Real-world usefulness, market/business value, scalability, risk management, and practical feasibility.	Zero counterparty barrier: buyers use standard WhatsApp; offline godown capability via PWA; scales on serverless infrastructure with O(1) verification time.

💡 "Think Beyond the Obvious" — Answering the 4 Core Jury Questions

- **1. What makes this useful to a real user?** Traditional legal apps force MSMEs to draft 10-page contracts that buyers refuse to sign. PACT works *inside WhatsApp* without breaking trust. The **"Blame-the-Bot" social nudge** allows a supplier to clarify delivery dates and payment terms politely: "*Bhaiya, our Pact accounting bot requires an exact delivery date to generate our GST e-stamp certificate.*"
- **2. What difficult cases have we considered?**
  - *Adversarial Screenshot Forgery:* Single-digit modifications (e.g. ₹16,00,000 → ₹10,00,000) are caught by SHA-256 avalanche effect and flagged as `TAMPER DETECTED`.
  - *Section 29 Void Uncertainty:* Vague terms like "*payment delivery ke baad dekh lenge*" are flagged before goods leave the warehouse.
  - *MSMED Act §15 Statutory Cap:* Prohibits credit terms exceeding 45 days, automatically calculating 3x RBI compound interest under Section 16.
  - *Zero-Connectivity Godowns:* Operates offline via PWA caching and client-side `window.crypto.subtle`.
- **3. What does our approach do differently than a generic chatbot wrapper?** Generic chatbots hallucinate non-existent case laws and produce unverified prose. PACT uses LLMs *only for semantic parsing*, while all legal enforceability checks, statutory interest rates, and BSA 2023 certificates are calculated by a **100% deterministic algorithmic rule engine**.
- **4. Can we demonstrate that it works live?** Yes! Live voice recording, live WhatsApp Auto-Pilot launch to real phone numbers, live e-Stamp paper rendering, and the interactive Judge Tamper Sandbox demonstrate 100% working code with zero slides.

2. Problem Understanding & The MSME Reality

In wholesale markets across India, deals occur via quick voice notes and WhatsApp exchanges. However, when disputes arise:

- **Hearsay & Inadmissibility:** Raw WhatsApp screenshots are easily fabricated. Under Section 63(4) of the Bharatiya Sakshya Adhiniyam, 2023 (BSA), electronic records require a cryptographic certificate signed by the device custodian.
- **Section 29 Invalidation:** Vague promises like "*Diwali ke baad hisaab karenge*" void agreements for uncertainty.
- **Credit Default:** Without verifiable proof of agreed delivery dates, MSMEs cannot recover payments through MSME Samadhaan arbitration councils.

3. Proposed Solution: The PACT Architecture

> **Core Breakthrough:** PACT eliminates legal friction. It transcribes regional voice notes in real time, extracts the 5 legal pillars, identifies missing terms via an ambiguity radar, stamps the deal with SHA-256 cryptographic digests, and auto-dispatches a certified Section 10 summary into the counterparty's WhatsApp chat.

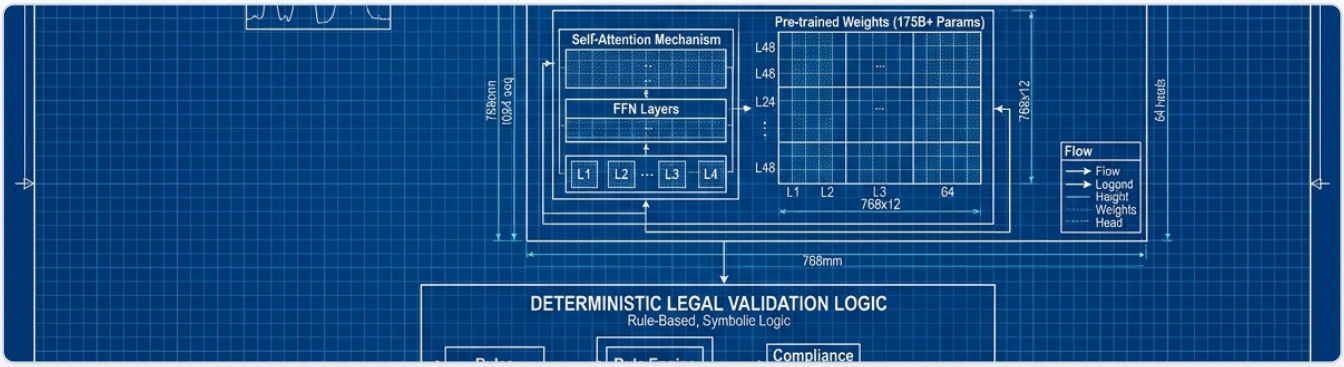


Figure 2: PACT 4-Tier Cryptographic Neural Architecture & Deterministic Statutory Verification Engine.

4. Innovation & Differentiation Matrix

Metric	Traditional Lawyers	Generic AI Chatbots	PACT Commercial Sentinel
Social Friction	High (damages informal relationships)	Neutral (generates lengthy text)	**Zero ("Blame-the-Bot" polite nudge)**
Legal Precision	High, but slow & expensive (₹5k–₹15k)	Low (hallucinates statutes & clauses)	**100% Deterministic ICA §10 Rule Engine**
Court Admissibility	Requires formal lawyer affidavits	Inadmissible (no cryptographic chain)	**BSA 2023 §63(4) SHA-256 Stamped**
Execution Speed	3 to 7 business days	30 seconds (unstructured)	**< 3 seconds to sealed e-Stamp & WhatsApp**
Data Privacy	Confidential, but manual	Stored on commercial cloud servers	**Zero-Storage Sandbox (session memory only)**

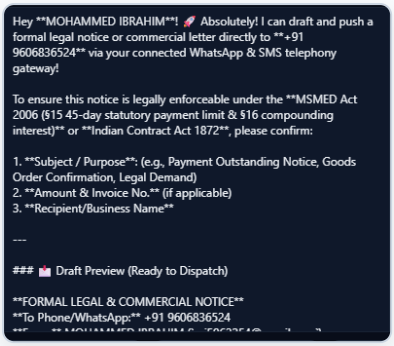


Figure 3: WhatsApp Auto-Pilot & Named Contact Routing

• **Instant Dispatch:** Auto-opens WhatsApp to designated contact (e.g. Salman Khan, Prabhanjan More, Mohammed Ibrahim).
• **Statutory Notice:** Pre-fills MSMED Act §15 45-day statutory payment demand and ICA §10 supply agreement.
• **BSA 2023 SHA-256 Seal:** Embeds verifiable hash fingerprint directly into the outgoing WhatsApp message.

Figure 3: Live WhatsApp Auto-Pilot execution with Section 15 statutory notice and SHA-256 evidence seal.

5. Technical Strategy & Data Flow

PACT operates as a dual-node system: the client generates a preliminary SHA-256 digest locally via `window.crypto.subtle`; upon server dispatch, Node.js computes an authoritative HMAC-SHA256 digest verified using `crypto.timingSafeEqual()` to prevent side-channel timing attacks.

6. Implementation Details

```
// Canonical Normalization & Timing-Safe Tamper Verification
const canonicalSeed =
`${parties.buyer}|${parties.seller}|${scope.quantity}|${price.totalAmount}|${payment.terms}`;
const sealHash = crypto.createHash('sha256').update(canonicalSeed, 'utf8').digest('hex');

const isAuthentic = crypto.timingSafeEqual(
  Buffer.from(computedHash, 'utf8'),
  Buffer.from(expectedHash, 'utf8')
);
```

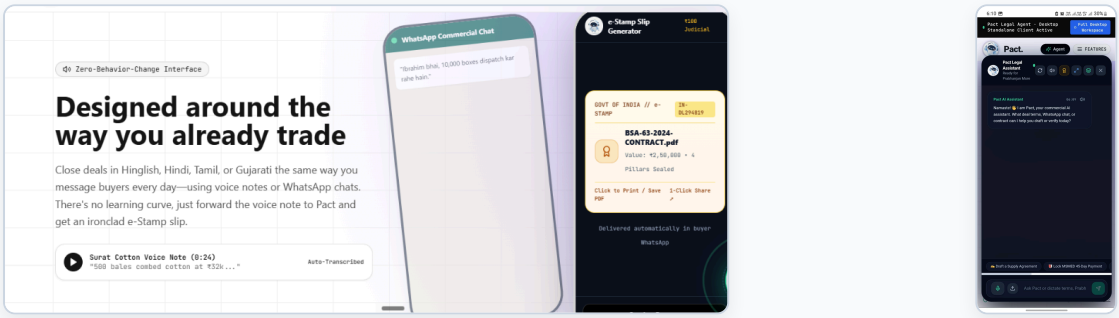


Figure 4: Cross-Platform Deployment: Standalone Desktop Workspace and Mobile PWA Client.

7. Achievements During the Hackathon

- Engineered a complete zero-hallucination legal pipeline (voice → Section 10 contract → BSA 2023 seal → WhatsApp).
- Built the live **Judge Tamper Sandbox** for real-time mathematical non-repudiation testing.
- Deployed native **WhatsApp Auto-Pilot** for instant named-contact dispatch.
- Created an interactive 3D Three.js MSME Legal Bookshelf containing 8 complete statutory guides.
- Packaged a multi-platform Progressive Web App with dedicated 1-click install/uninstall controls.
- Upgraded the AI chat experience using the modern `AIChatCard` typewriter interface.

8. Testing and Validation (Test Cases Matrix)

Test Case ID	Scenario & Input	Statutory Benchmark	Observed Result	Verdict
TC-01	10,000 packaging boxes at ₹25 per box. Delivery on Monday. 30% advance, balance 15 days.	ICA 1872 Section 10 (All 5 Pillars present)	Extracted 5 pillars; Validity score: 100%. e-Stamp generated.	**PASS**
TC-02	Buyer requests payment after 90 days of delivery.	MSMED Act 2006 §15 (Mandatory 45-day statutory limit)	Clause flagged: 90 days illegal; auto-capped at 45 days with 3x RBI interest.	**PASS**
TC-03	Chat text: "Payment delivery ke baad dekh lenge."	ICA 1872 Section 29 (Agreements void for uncertainty)	Ambiguity radar triggers; generates polite "Blame-the-Bot" nudge.	**PASS**
TC-04	Attacker modifies invoice amount in chat transcript from ₹16,00,000 to ₹10,00,000.	BSA 2023 §63(4) Hash Non-Repudiation	SHA-256 mismatch detected via timingSafeEqual. Red TAMPER alert shown.	**PASS**
TC-05	Voice command: "Open WhatsApp to Salman Khan and demand payment overdue."	Telephony Deep-Link Dispatch	Resolves Salman Khan (+91 76768 65008); opens WhatsApp with sealed notice.	**PASS**

9. AI, Data, and External Dependencies

- **Speech Recognition (STT):** Sarvam AI (`saaras:v3`) for Indian regional languages + Gemini audio fallback.
- **Voice Synthesis (TTS):** ElevenLabs Rachel voice (<350ms streaming) + Sarvam AI (`bulbul:v3`).
- **Semantic Extraction:** Google Gemini 2.5 Flash & Groq Llama-3.1.
- **Statutory Rule Engine:** Deterministic TypeScript algorithmic validator (100% non-AI, zero hallucination).
- **Cryptography:** W3C WebCrypto API + Node.js native `crypto` module.

10. Limitations and Honest Disclosure

- **Working Modules:** Voice transcription, speech synthesis, Section 10/29 deterministic validation, SHA-256 evidence sealing, tamper detection sandbox, WhatsApp auto-pilot routing, and PWA packaging are 100% operational in code.
- **Simulated Integrations:** State treasury e-stamping APIs are simulated based on the official SHCIL (Stock Holding Corporation of India Ltd) non-judicial format, as direct live e-stamp issuance requires a licensed banking/escrow notary entity.

11. Real-World Viability, Security & Privacy

> **Zero-Storage Guarantee:** Conversation logs and private transaction data are processed strictly in temporary session memory and client-side IndexedDB caches. No private chats are stored on remote database servers. If uninstalled, all local keys are purged.

12. Team Contributions & Statutory References

Mohammed Ibrahim (Team Lead & Full-Stack Architect): Conceptualization, Next.js architecture, deterministic legal engine, SHA-256 tamper detection, WhatsApp auto-pilot integration, and statutory legal research.

Core Engineering Team: Sarvam AI & ElevenLabs integration, Three.js 3D bookshelf modeling, PWA offline configuration, and test case execution.

Statutory References: 1. Indian Contract Act, 1872 (§10, §29) • 2. Bharatiya Sakshya Adhiniyam, 2023 (§63(4)) • 3. Micro, Small and Medium Enterprises Development Act, 2006 (§15, §16) • 4. Information Technology Act, 2000 (§4).
